



- 6** The plane Π has equation $x + 3y + 2z = 1$.

- (a) Find the perpendicular distance from the origin O to the plane Π . [2]

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Relative to O , the points A, B, C have position vectors

$$-\mathbf{j} + 2\mathbf{k}, \quad 2\mathbf{i} - \mathbf{k}, \quad 2\mathbf{i} - \mathbf{j} - \mathbf{k},$$

respectively.

- (b)** Find the acute angle between the planes OAB and Π . [4]

[illegible]



(c) Find an equation for the common perpendicular to the lines OC and AB .

[8]

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